

MiniMax the idea

For as long as I can remember I've had the idea to produce the smallest practical rc yacht. With modern electronics and 3D printing the barrier to building something like this is removed and it is entirely possible to accurately build a tiny r/c yacht that performs well.

Over the last 25 years I did some extensive work in the Footy class learning along the way what is important in very small yachts and developing some new ideas. The Footy as nice as it is is typically not able to be built in 1 piece on a typical home 3d printer. The yachts have quite deep draft and large rigs also.

I wanted a truly small yacht I could carry easily, sail in small shallow waterways and hopefully inspire people to take their first steps with radio sailing.

I think I have achieved that with MiniMax.

There are downsides though..the yacht is cheap and fairly easy to build by radio sailing standards..BUT, as yachts get smaller they do become harder to sail well, they feel every wind shift, suffer from the effects of wind friction at surface level and operate at Reynolds numbers which mean the hull thinks its sailing through a thick syrup! Also at only 120 grams displacement it carries no inertia what's so ever through tacks!. So be prepared for all those challenges. That said MiniMax is a cool wee yacht when trimmed well and will give hours of fun.

It will even be more fun when several or more MiniMaxs are sailing together.

I suggest a very informal rule set be used when racing as the yachts can be very unpredictable and hard to see in close quarters. They are small and light so they can bounce off each other..collisions are costly in race situations for both parties..so no right of way rules is the best way to race these yachts in my opinion.

Beginners could start with reaching courses and even shore starts, the group can develop its skills and progress to more advanced race courses.

I haven't presented any class rules as such yet. I have ideas and concepts in mind if the little thing becomes popular and racing starts. maybe it will maybe it won't and the majority just want a little yacht to sail around or a cool little sailboat that can sit fully rigged in your office!

Building MiniMax is as straightforward as I can make it with several innovative features which make it easy.

The ballast is a portion of a 12mm s/s bolt available from any hardware shop anywhere in the world.

We have specced the tiny 2 gram generic servos that are very cheap, paired with a balanced una Mac rig the sheeting loads are small and this all helped keep the weight down. Building MiniMax is a project more like building an indoor rc plane than your typical model yacht. You will likely need tweezers to do some of the work and I actually use them on sailing days to take the battery in and out. Be very careful of excessive weight gain. MiniMax is quite minimalist and needs to be to work effectively.

Lastly, it is my intention to produce MiniMax kits for those who cannot do the printing themselves and also if momentum grows the possibility of licensing builders in other countries is possible.

Sailmakers can produce MiniMax sails etc and a whole community can develop.

Have fun on your MiniMax adventure..good print settings and fairwinds!

Brett McCormack

